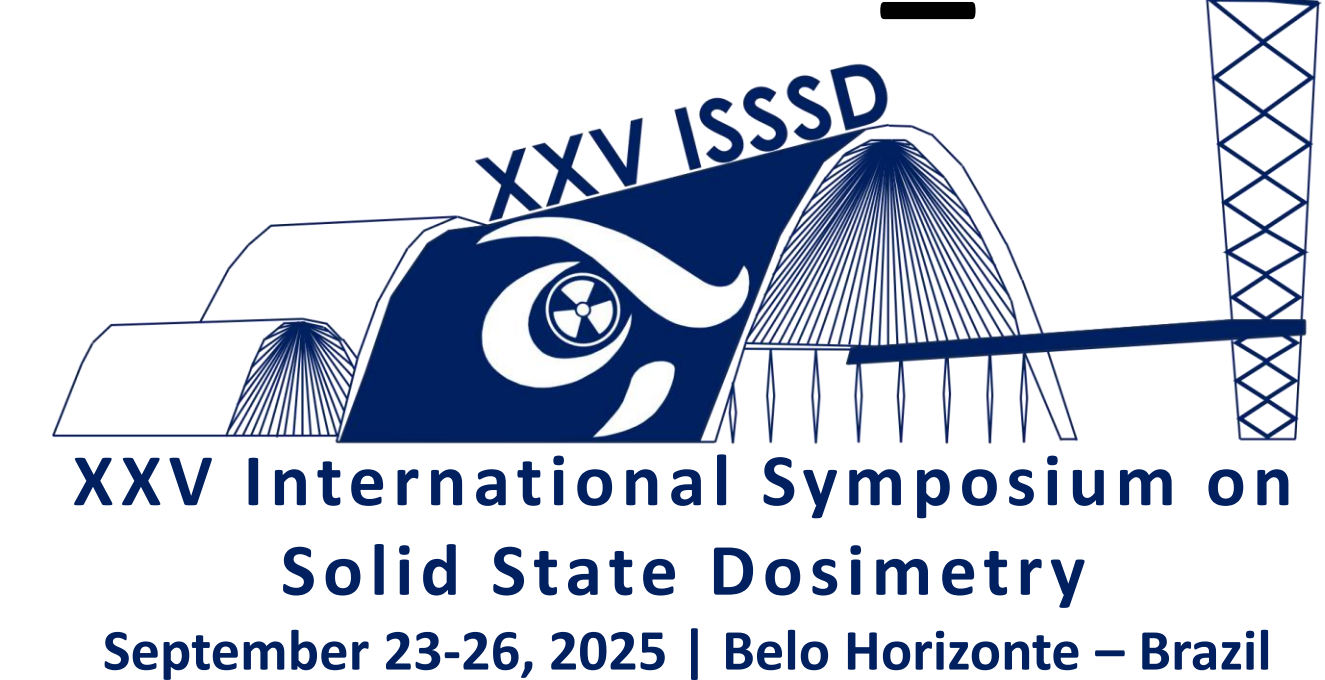


Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose_X Monte Carlo Simulations

P. Vosiak^{1*}, A. Yagui^{1,2}, H. R. Schelin^{1,2}, P. Castilho, R. Mello¹, S. Paschuk², M. R. de S. Roberto³, A. C. Da Silva³, V. Asfora³ and A. Legnani²

¹Pelé Pequeno Príncipe Research Institute; ²Federal University of Technology – Paraná; ³Federal University of Pernambuco

*Correspondence to: paula.vosiak@gmail.com



1. INTRODUCTION

The increasing use of radiological examinations in pediatrics raises concerns about ionizing radiation exposure, particularly in radiosensitive organs. Children are more vulnerable due to their smaller body size and longer life expectancy, which increases the risk of stochastic effects such as radiation-induced cancer. Abdominal and pelvic radiographs are among the most frequently performed examinations in this age group, often involving critical organs such as the bladder, gonads, and bone marrow. This study aimed to estimate organ absorbed doses in pediatric radiographs and discuss the application of age-specific Diagnostic Reference Levels (DRLs).

2. MATERIALS AND METHODS

- Design: cross-sectional study in a large pediatric hospital in Brazil.
- Age groups: 0–1, 1–5, 5–10, 10–15, ≥ 15 years.
- Equipment: two Shimadzu digital systems dedicated to pediatrics.
- Parameters recorded: age, sex, weight, height, kVp, mAs, focus-skin distance (FSD), $K_{a,i}$, PKA.
- Tool: Caldose_X Monte Carlo simulations.
- Organs evaluated: liver, spleen, intestines, bladder, gonads, bone marrow.

3. RESULTS

- The absorbed dose estimates showed that the most exposed organs during pediatric abdominal and pelvic examinations were the bladder, gonads, intestines, spine, and bone marrow, reflecting their anatomical location relative to the primary X-ray field.
- It was observed that absorbed dose increased progressively with patient age, following body growth and the need for higher technical parameters to maintain diagnostic image quality. For example, skin entrance dose rose from approximately 0.26 mGy in infants (0–1 year) to 1.07 mGy in adolescents (≥ 15 years), while liver dose increased from 0.11 mGy to 0.25 mGy across the same age groups.
- In addition, wide intra-group variability was observed, attributed to individual anatomical differences, patient positioning, and the extent of collimation. This variability was particularly evident in peripheral organs such as the gonads, where small changes in the irradiated field size resulted in significant dose differences.
- Statistical analysis (ANOVA) revealed highly significant differences ($p < 0.001$) in dose distribution among the pediatric age groups, for both abdominal and pelvic examinations, highlighting the direct impact of age on exposure patterns.
- The results were consistent with international literature, such as the studies by Nahangi & Chaparian (2015), Kiljunen et al. (2009), and Lacerda et al. (2008), which also demonstrated a progressive increase in dose with age and emphasized the importance of protocol standardization.
- These findings confirm the relevance of using tools such as Caldose_X, which enable individualized and more realistic organ dose estimates, overcoming the limitations of methods based solely on general physical quantities such as $K_{a,i}$ or PKA.

4. CONCLUSIONS

- The study confirmed that the bladder, gonads, intestines, bone marrow, and liver are among the most exposed organs in pediatric abdominal and pelvic examinations, with absorbed doses progressively increasing with age.
- Intra-group variability reinforces the need for individualized protocols and strict collimation to reduce unnecessary exposure.
- The findings support the implementation of age-specific DRLs in Brazil and highlight the importance of optimization strategies for pediatric radiological protection.
- Multicenter studies are recommended to validate these results and contribute to the establishment of national benchmarks.

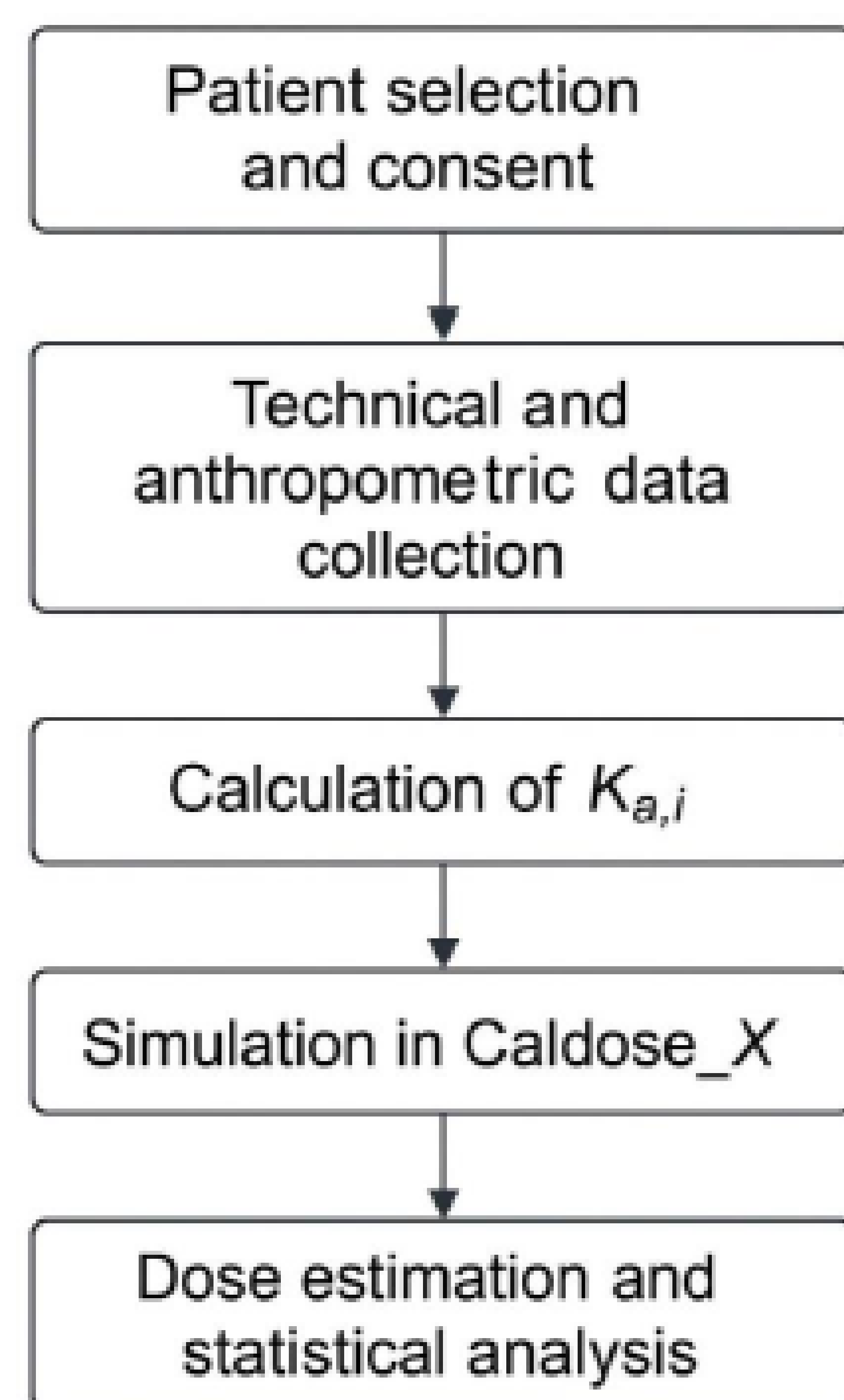


Figure 1 shows methodological workflow from patient data acquisition to organ dose estimation using Caldose_X.

ACKNOWLEDGMENTS

The authors thank CAPES, CNPq, CNEN, and Fundação Araucária for financial support.